DESIGN AND IMPLEMENTATION OF WIRED/WIRELESS NETWORK USING CISCO PACKET TRACER

Table of Contents

1. Introduction....4

2. Network Requirements and Topology Design....4

2.1 Understanding the Functional Requirements....4

2.2 Designing the Physical and Logical Topology....5

3. IP Addressing and Subnetting....6

3.1 Generating the Base Network Block from the Student ID....6

Block IP:....6

191.51.99.0 /25 Range: 191.51.99.0 – 191.51.99.127 (128 total addresses)....6

3.2 Step-by-Step Subnet Calculation Process....6

4. Device Configuration and Logical Implementation....8

4.1 Router Configuration and Interface Assignments....8

4.2 Switch and Host Configuration....10

5. Wireless Network Configuration....12

5.1 SSID Broadcasting and Security Implementation....12

5.2 Wireless Laptop Configuration and Connectivity Testing....13

6. Routing and End-to-End Communication....15

6.1 RIP Routing and Route Propagation....15

6.2 Connectivity Testing and Interpretation....17

7. Conclusion....23

References....24

Appendix:....27

List of Figures

Figure 1 Overview of network topology 5
Figure 2 R1: Configuring LAN and serial interfaces 9
Figure 3 R2: Configuring LAN and Serial Interfaces 9
Figure 4 R2: Configuring LAN and serial Interfaces 10
Figure 5 Subnet B Host: Assigning static IPs 11
Figure 6 Subnet C host: Assigning Static IPs 11
Figure 7 Subnet A Host: Assigning Static IPs 12
Figure 8 Wireless Subnet: Assigning SSID and WPA2-PSK security to Access Point 13
Figure 9 Wireless Laptop Configuration: Assigning Static IPs 14
Figure 10 Wireless Configuration: Connecting laptop to access point using passkey 14
Figure 11 Subnet A: Wireless connection established 15
Figure 12 R1: RIP V2 configuration 16
Figure 13 R2: Configuring RIP V2 16
Figure 14 R3: Configuring RIP v2 17
Figure 15 Ping from A1 to gateway, Subnet B Host and subnet C host: connectivity failing initially 18
Figure 16 Inter subnet connectivity test: Ping from A1 to gateway, Subnet B Host and subnet C host (Ping successful) 18
Figure 17 Checking Connectivity within the subnet: HostB1 to HostB2 (Connection successful) 19
Figure 18 Simulation mode: Ping from ServerC to LaptopA3 19
Figure 19 Simulation mode: Ping from ServerC to LaptopA3 (Sending Packets) 20
Figure 20 Simulation mode: Ping from ServerC to LaptopA3 (PDU Information: Sending out the frame) 20
Figure 21 Simulation mode: Receiving packet back 21
Figure 22 Simulation mode: Server C receives the frame 21
Figure 23 Ping from ServerC to LaptopA3: Packets sent and received successfully 22

1. Introduction

Modern computer networks have been designed and implemented with wired and wireless technology in combination to facilitate communication among various forms of devices. The aim of this coursework was to create a working network within Cisco Packet Tracer which will connect wireless laptops, wired hosts, servers, routers, and switches. However, before delving into technical issues like routing protocols, subnetting, and configuration of interfaces, it is prudent to have a clue on the simple principles of wireless networking, which is an essential part of the assignment (Mwansa *et al.*, 2024).

Wireless connectivity is based on radio frequency signals instead of cables and as such, it needs some means of good identification and secure access. The SSID (Service Set Identifier) is one of such mechanisms and this is the unique broadcast name of a wireless network. The SSID enables the wireless devices to identify and connect to the appropriate access point (Bahri *et al.*, 2025). Each wireless network should have an SSID and it should be easily identifiable against the other networks in the vicinity to prevent misuse and establish a stable connection.

The other theme in the wireless networking is WPA2-PSK (Wi-Fi Protected Access 2 Pre-Shared Key) which encrypts and authenticates wireless connections. WPA2 also keeps the network closed to the unauthorised users with the wrong wireless password. Encryption holds particular significance due to the fact that wireless signals are by their nature susceptible to interception due to their open air nature as opposed to wired traffic. Use of a robust pre-shared key will help to avoid unauthorised access, safeguard data and integrity of the network (Briones *et al.*, 2021). This is because these wireless concepts form the basis on which the whole network design is based.

2. Network Requirements and Topology Design

2.1 Understanding the Functional Requirements

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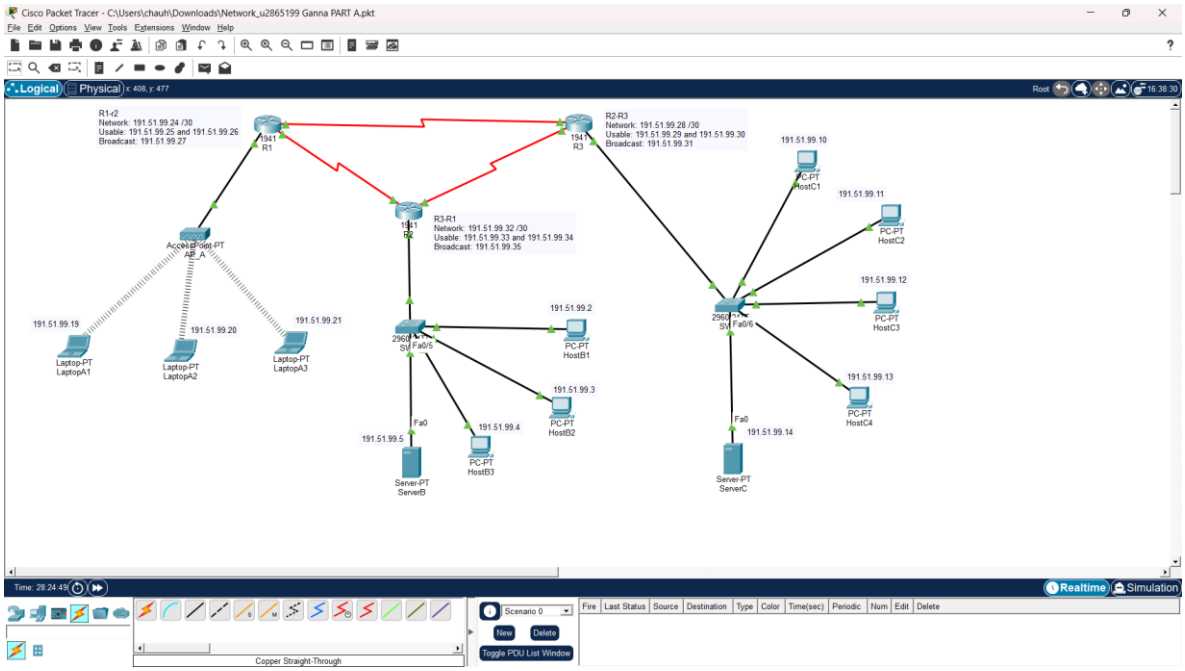
Cisco Packet Tracer - C:\Users\chauh\Downloads\Network_u2865199 Ganna PART A.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
R1-r2
Network: 191.51.99.24 /30
Usable: 191.51.99.25 and 191.51.99.26
Broadcast: 191.51.99.27
R2-R3
Network: 191.51.99.28 /30
Usable: 191.51.99.29 and 191.51.99.30
Broadcast: 191.51.99.31
R3-R1
Network: 191.51.99.32 /30
Usable: 191.51.99.33 and 191.51.99.34
Broadcast: 191.51.99.35
R1
R3
191.51.99.19
191.51.99.20
191.51.99.21
Laptop-PT LaptopA1
Laptop-PT LaptopA2
Laptop-PT LaptopA3
191.51.99.2
PC-PT HostB1
191.51.99.3
PC-PT HostB2
191.51.99.4
PC-PT HostB3
191.51.99.5
Server-PT ServerB
191.51.99.10
PC-PT HostC1
191.51.99.11
PC-PT HostC2
191.51.99.12
PC-PT HostC3
191.51.99.13
PC-PT HostC4
191.51.99.14
Server-PT ServerC
Fa0/5
Fa0/6
Fa0
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Copper Straight-Through

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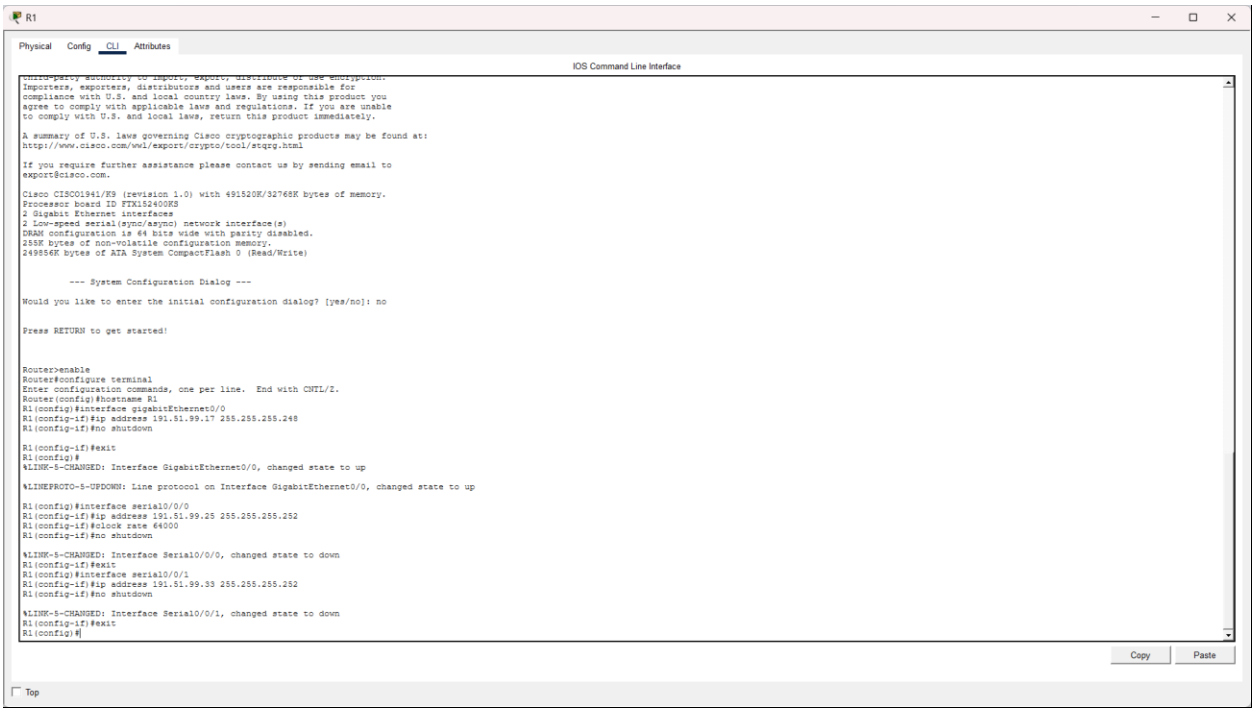

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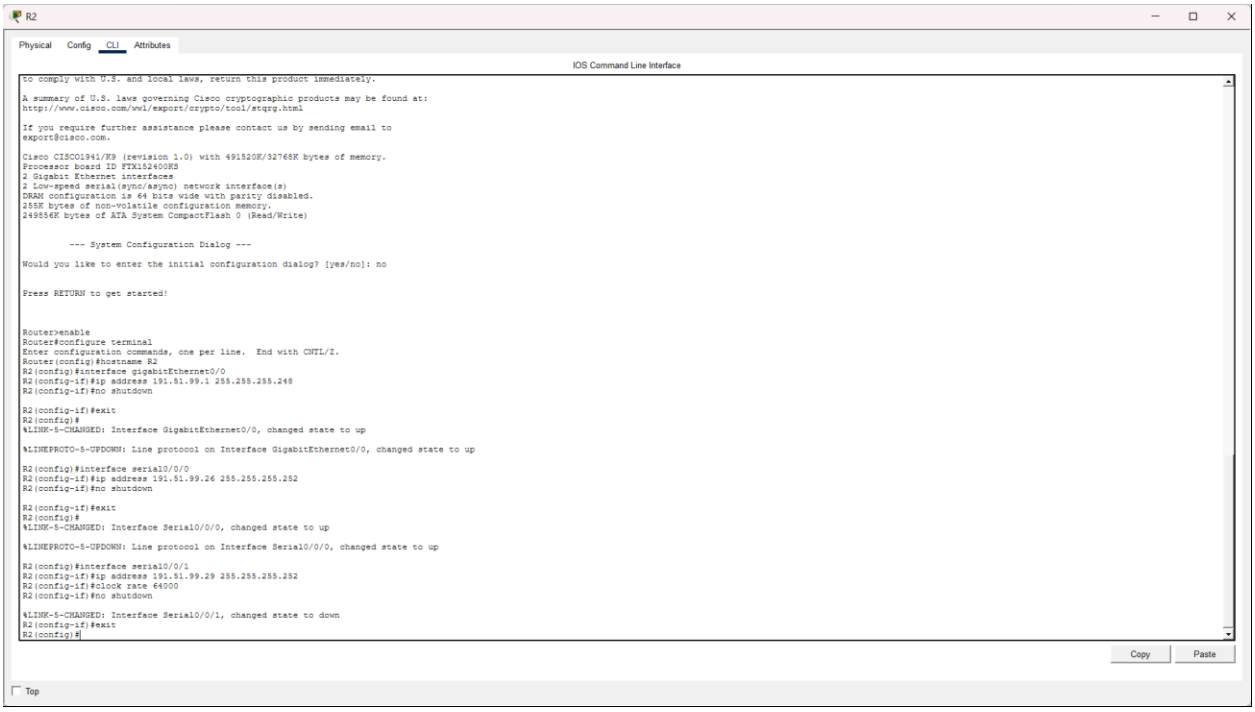

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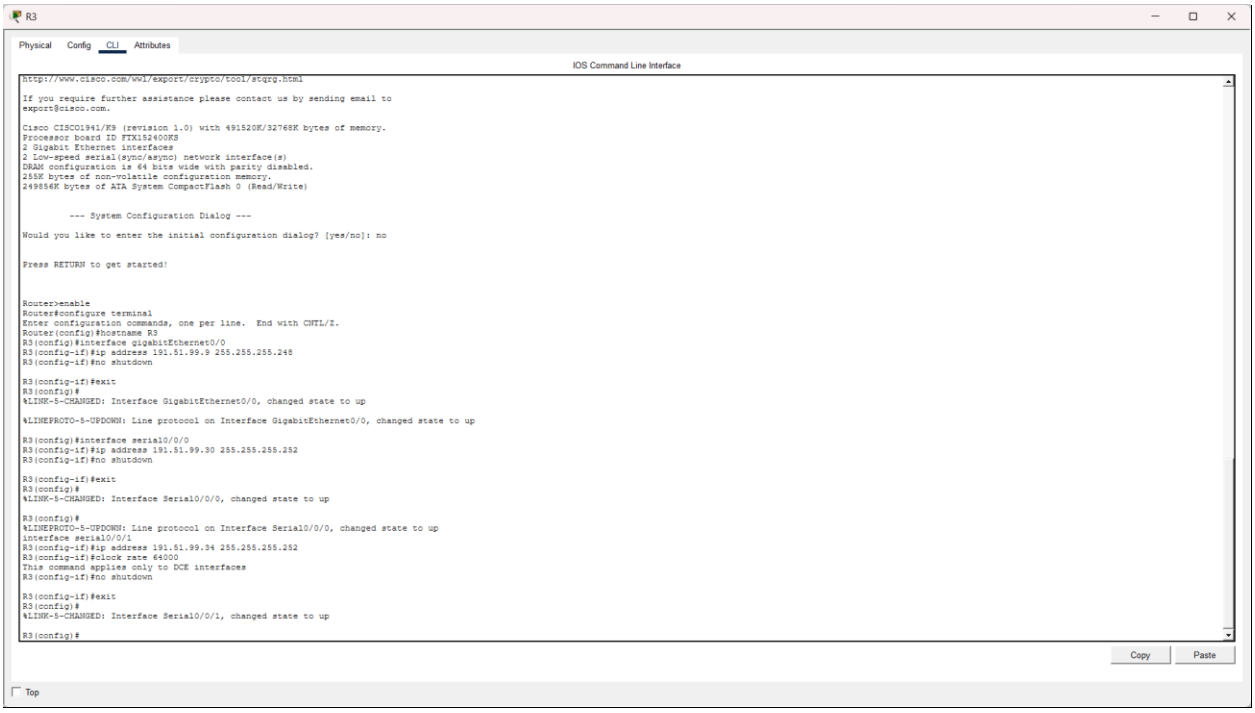

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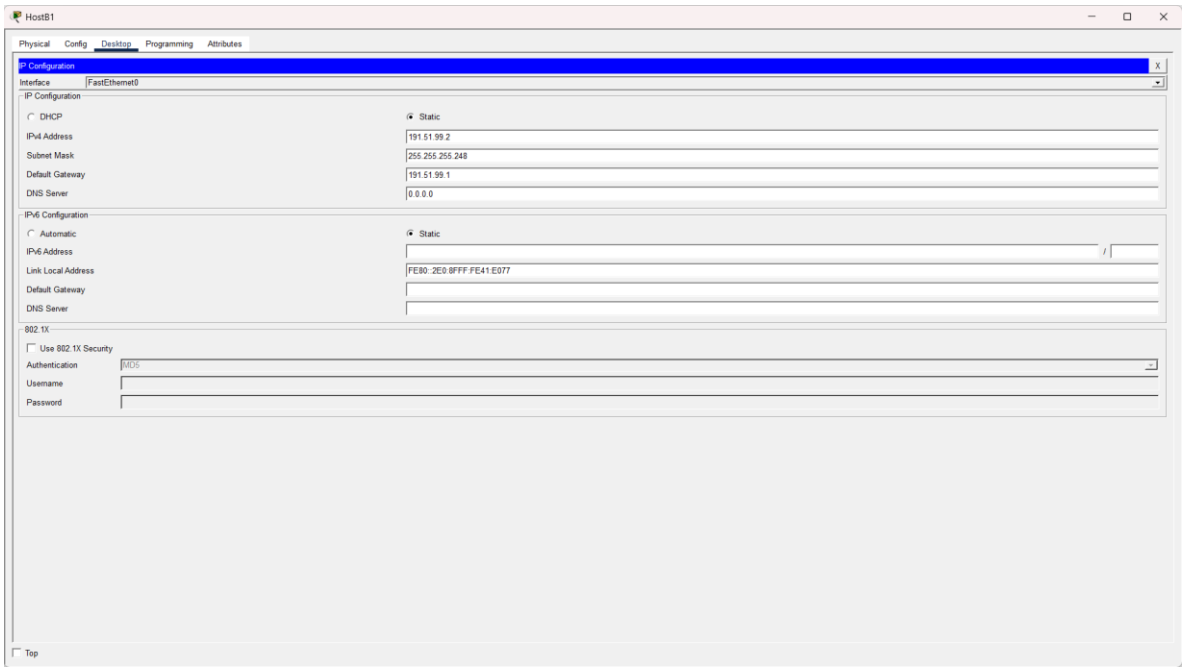

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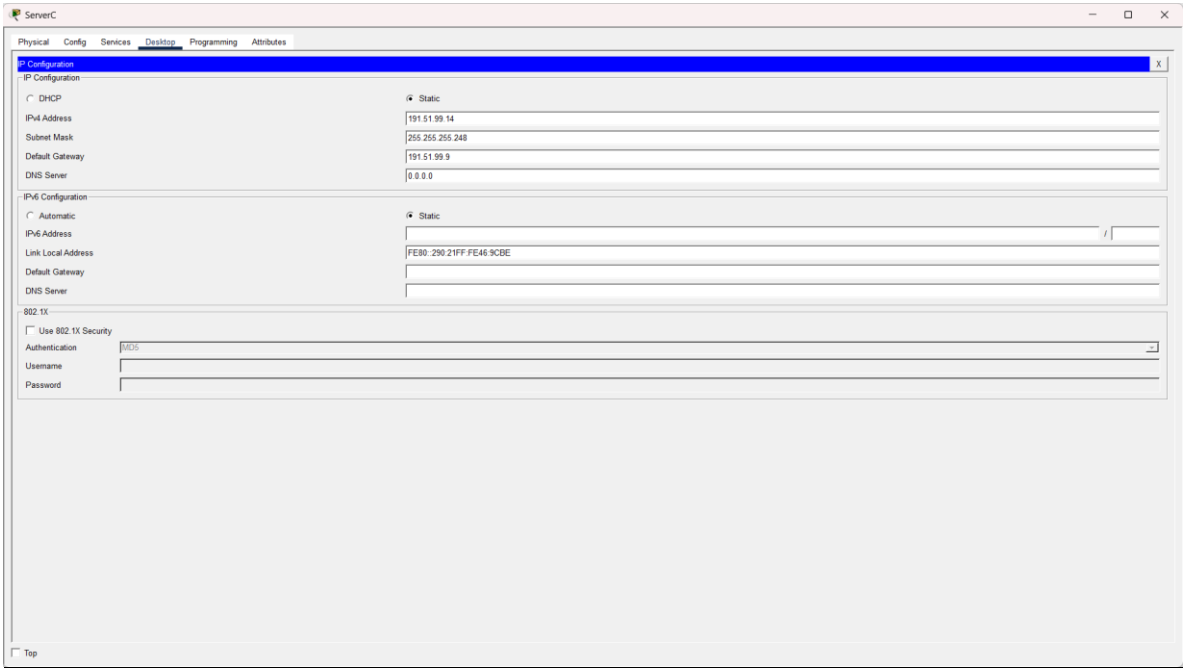

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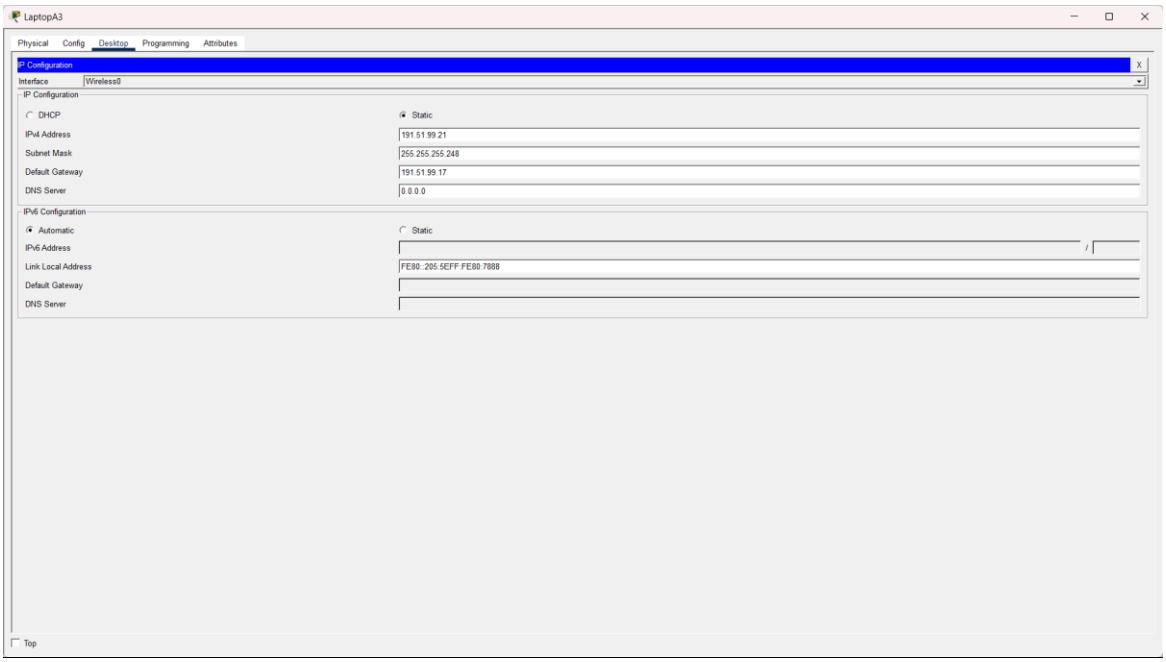

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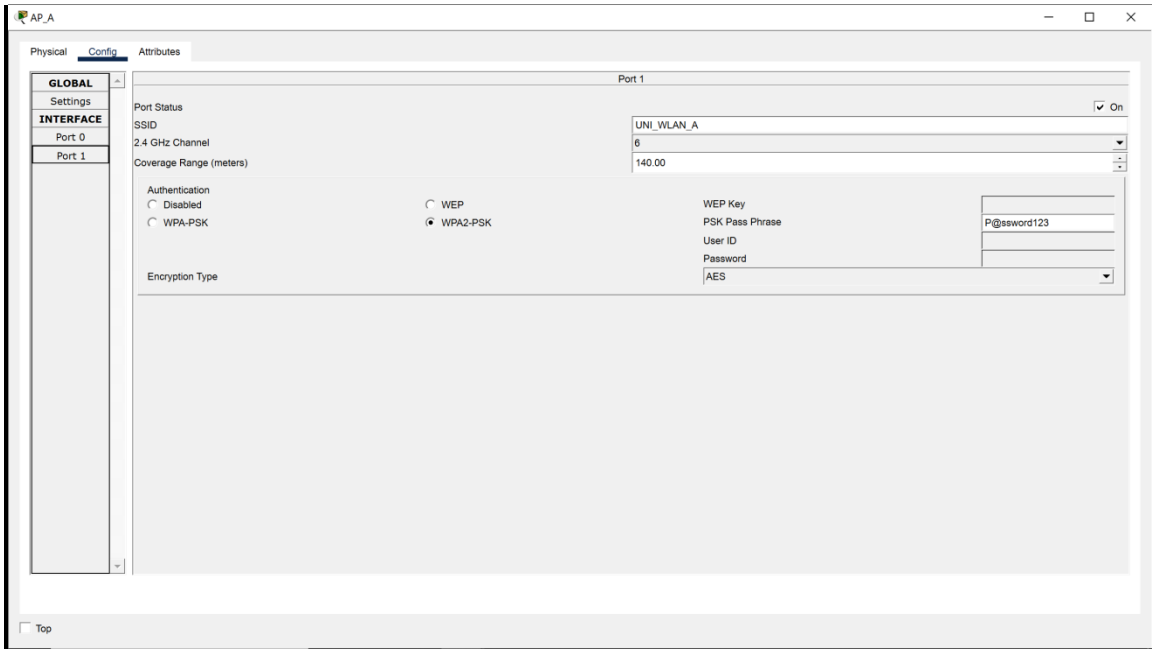

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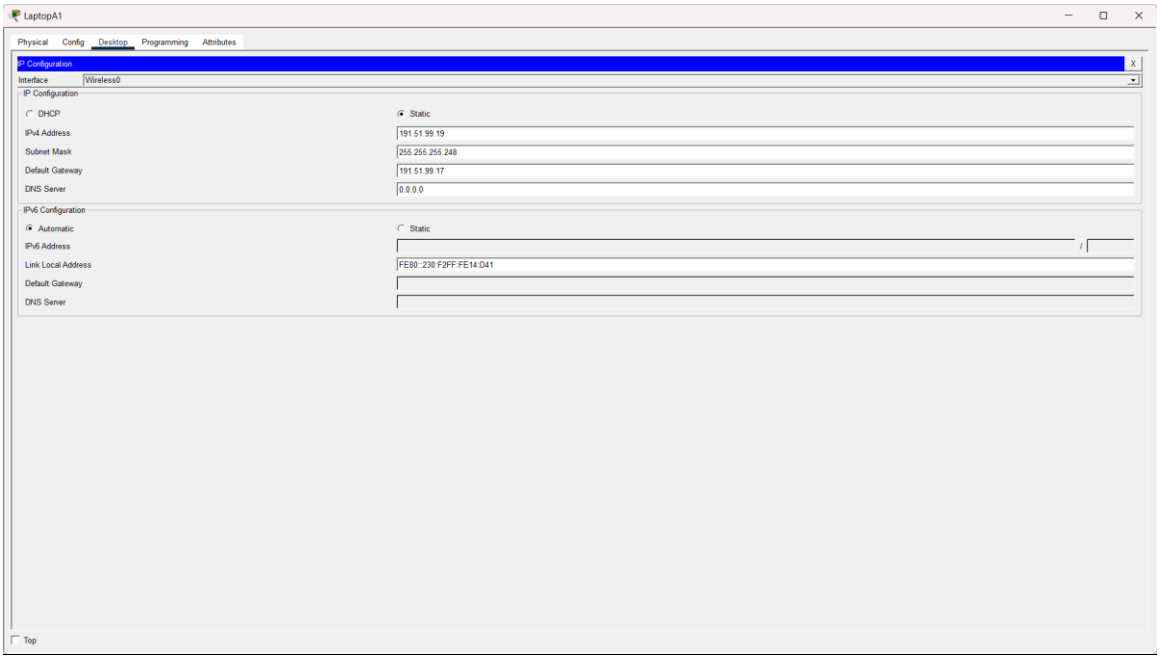

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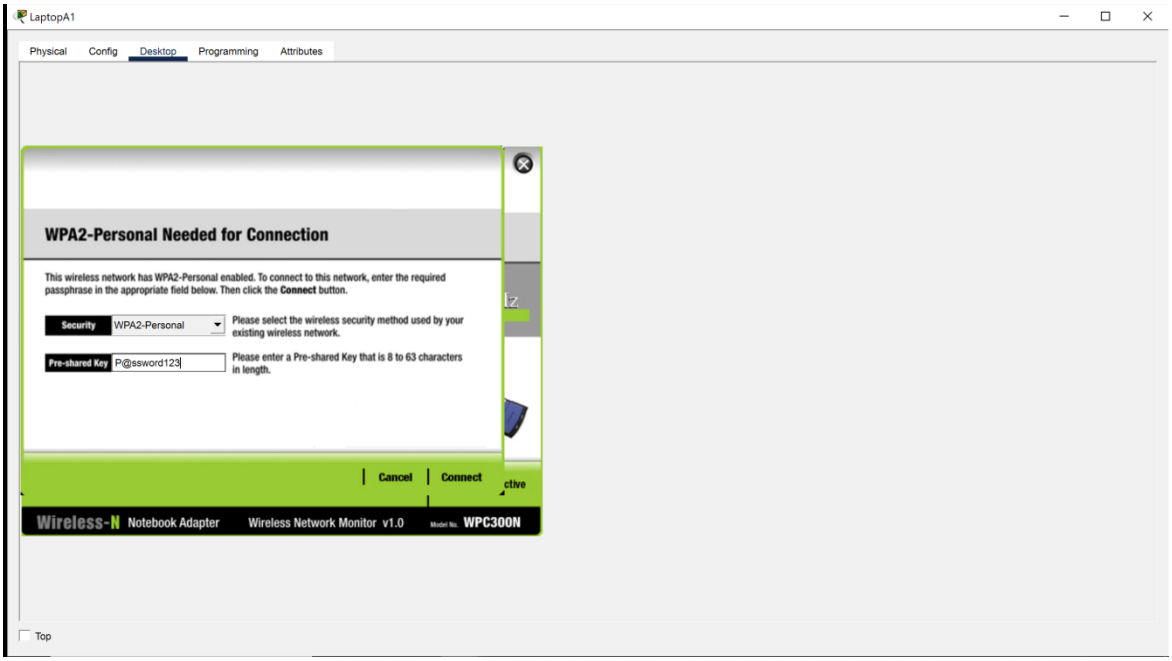

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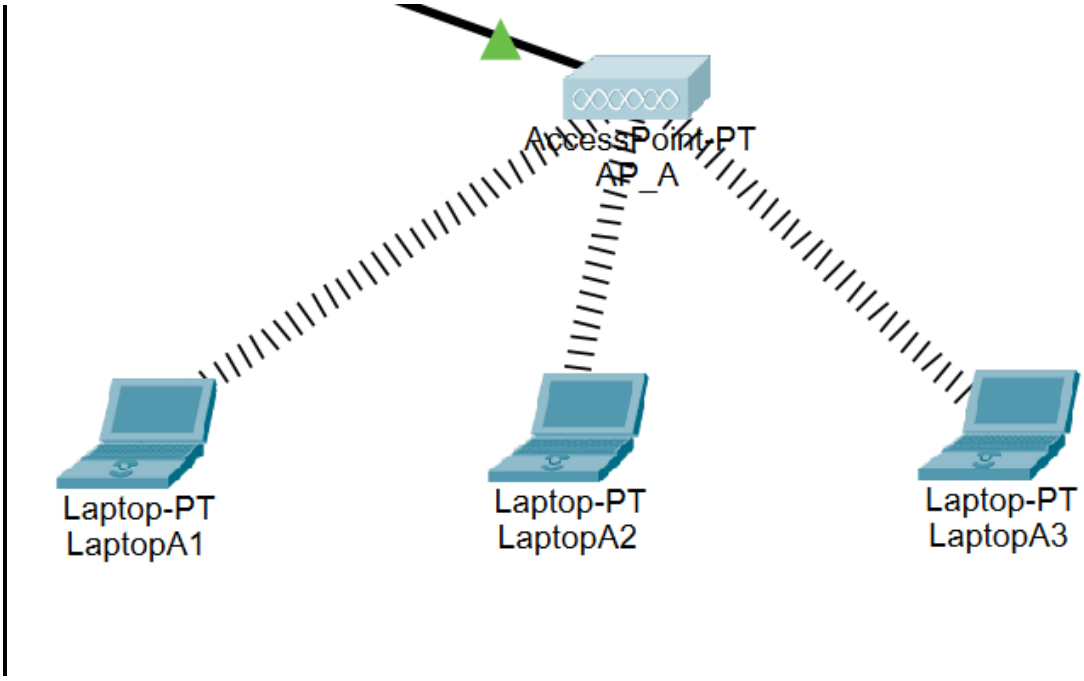

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R1

Physical Config CLI Attributes

IOS Command Line Interface

```
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface gigabitEthernet0/0
R1(config-if)#ip address 191.51.99.17 255.255.255.248
R1(config-if)#no shutdown

R1(config-if)#exit
R1(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config)#interface serial0/0/0
R1(config-if)#ip address 191.51.99.25 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#exit
R1(config)#interface serial0/0/1
R1(config-if)#ip address 191.51.99.33 255.255.255.252
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R1(config-if)#exit
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 191.51.99.0
R1(config-router)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

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Top

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R2

Physical Config CLI Attributes

IOS Command Line Interface

```
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface gigabitEthernet0/0
R2(config-if)#ip address 191.51.99.1 255.255.255.248
R2(config-if)#no shutdown

R2(config-if)#exit
R2(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config)#interface serial0/0/0
R2(config-if)#ip address 191.51.99.26 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#exit
R2(config)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R2(config)#interface serial0/0/1
R2(config-if)#ip address 191.51.99.29 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R2(config-if)#exit
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 191.51.99.0
R2(config-router)#exit
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

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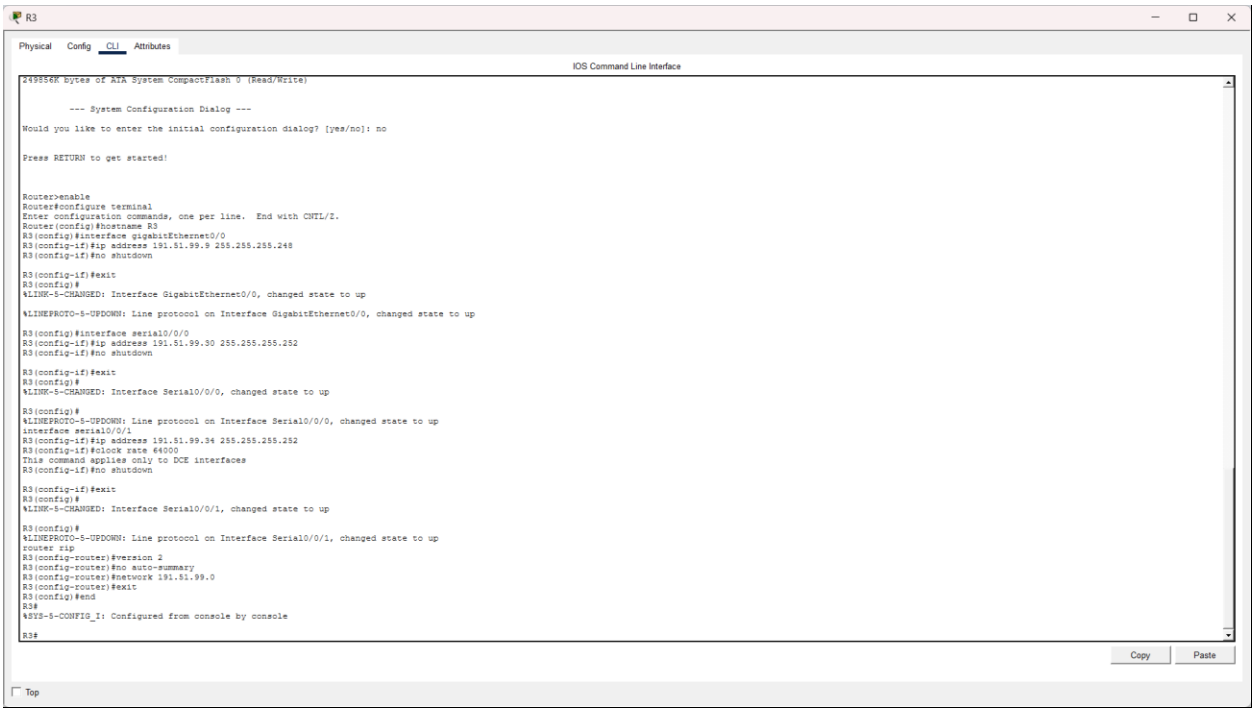

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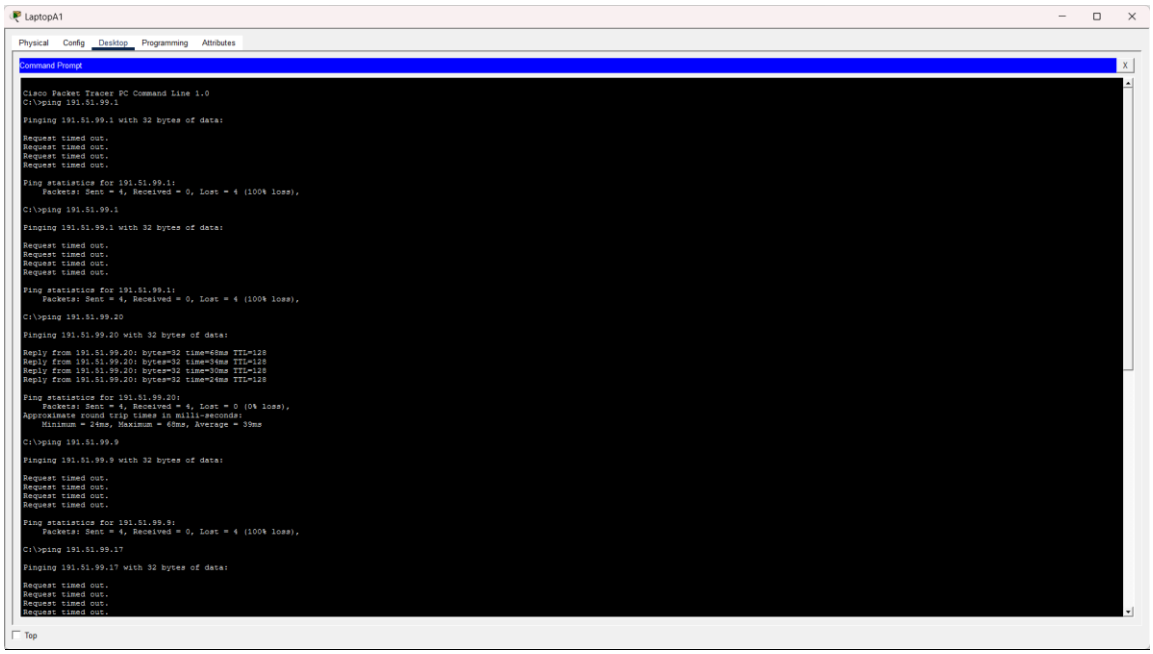

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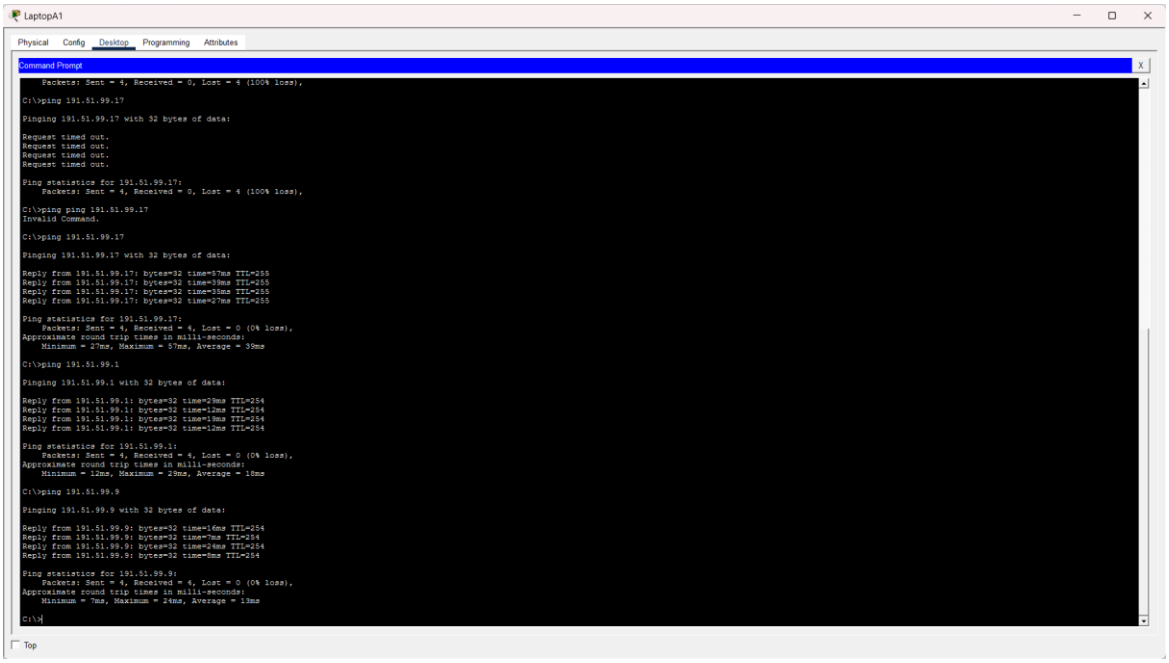

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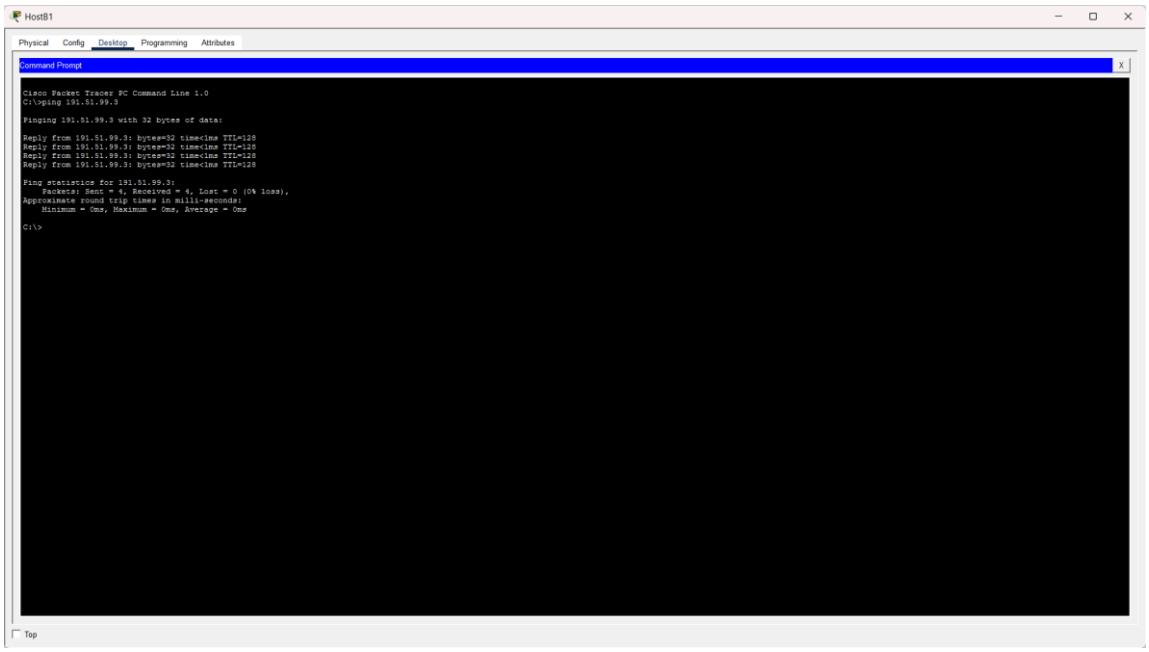

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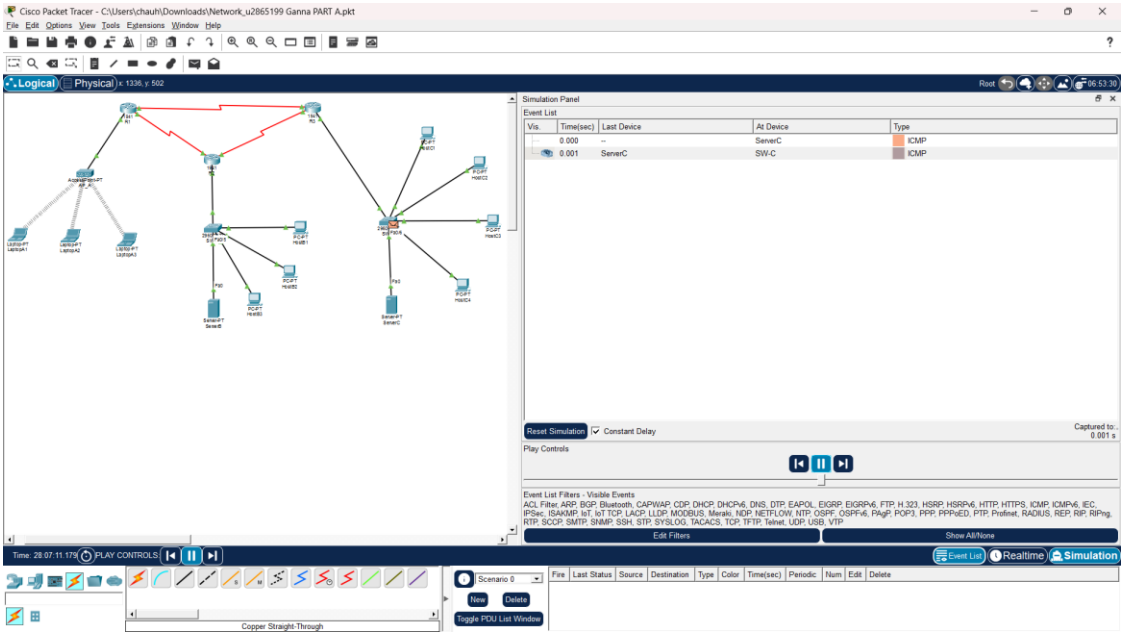

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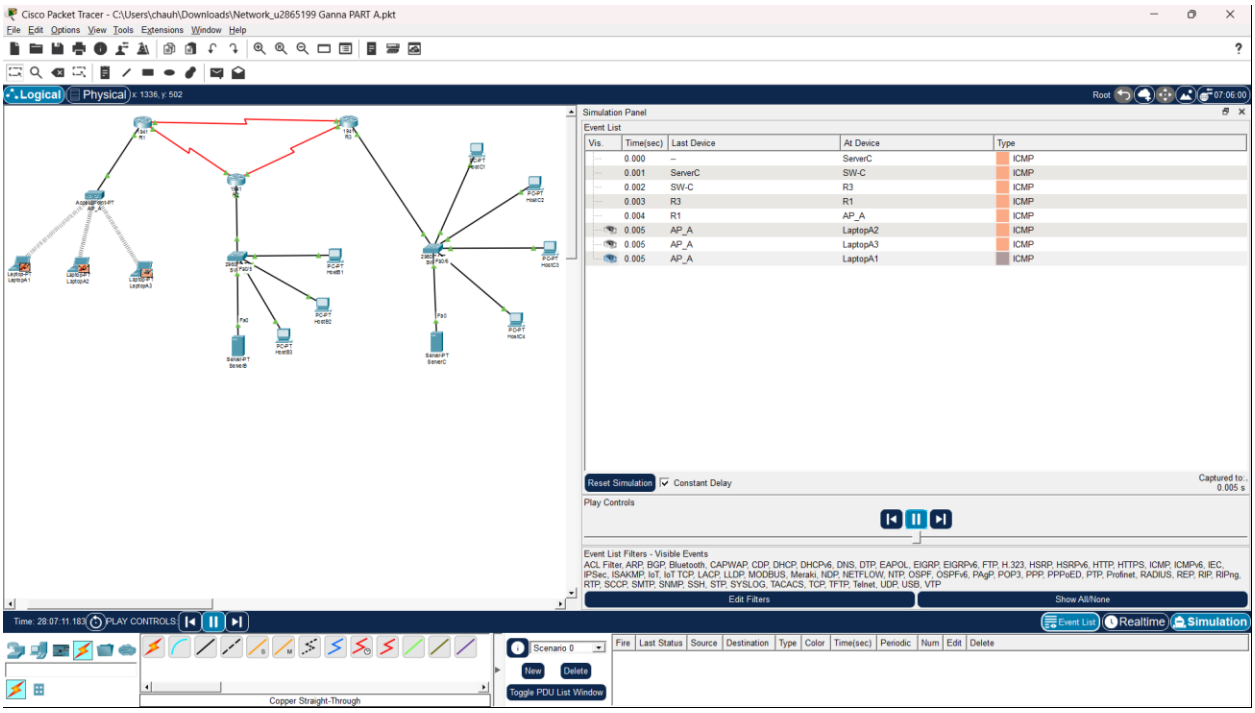

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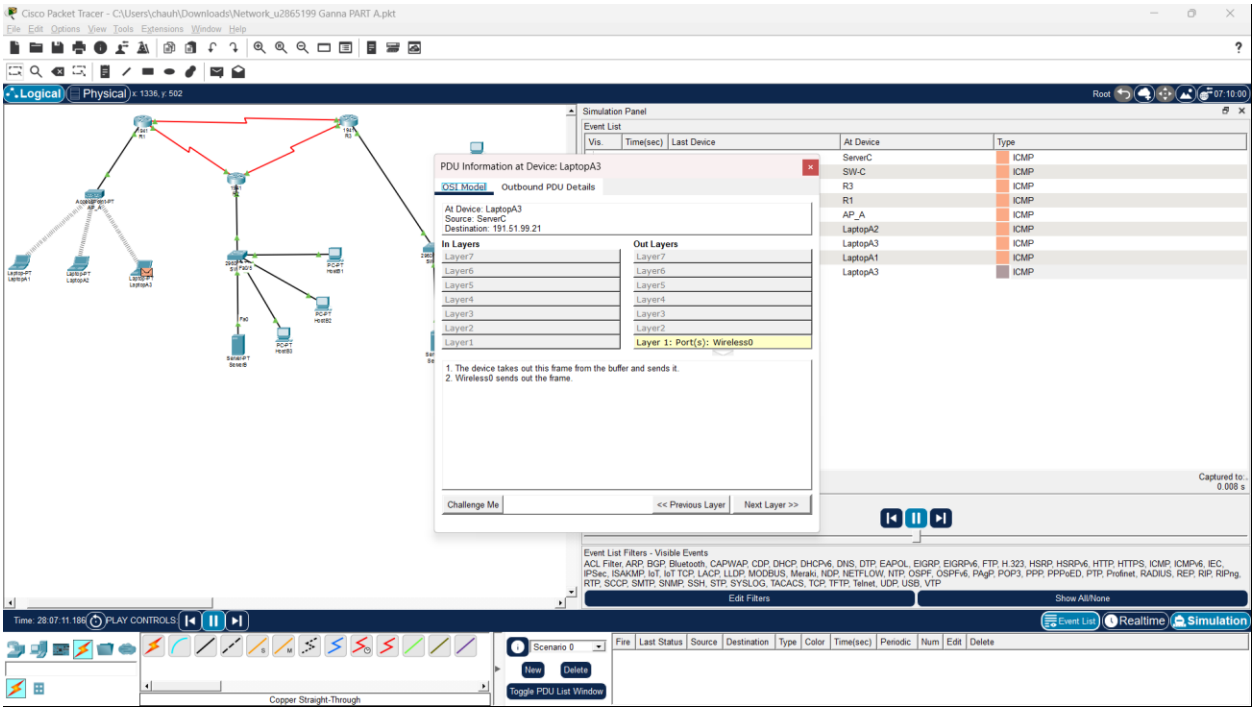

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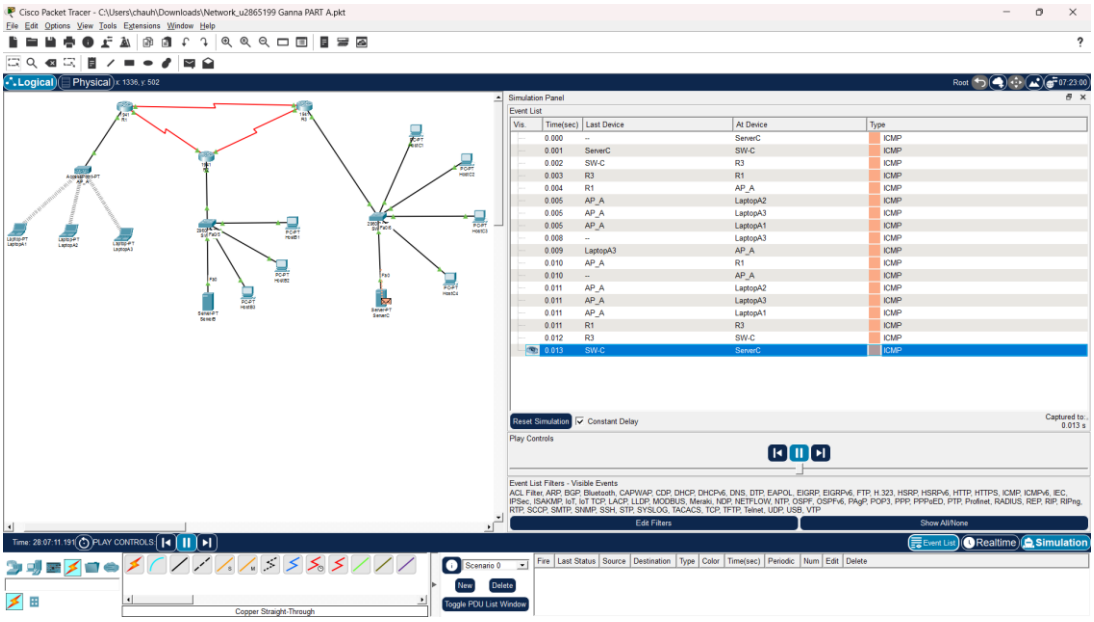

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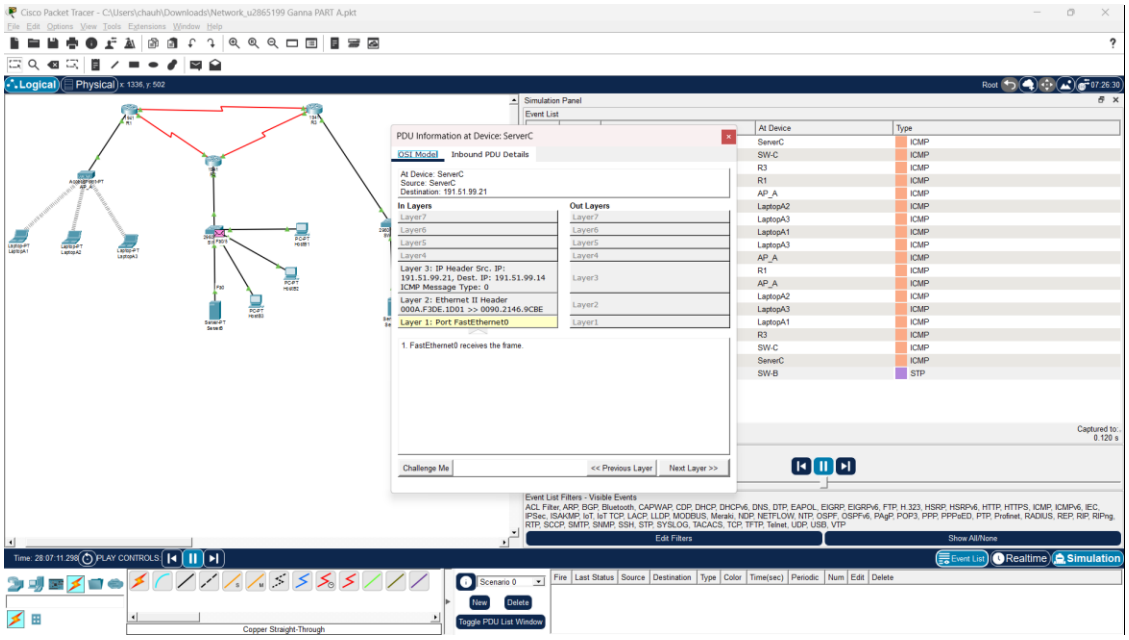

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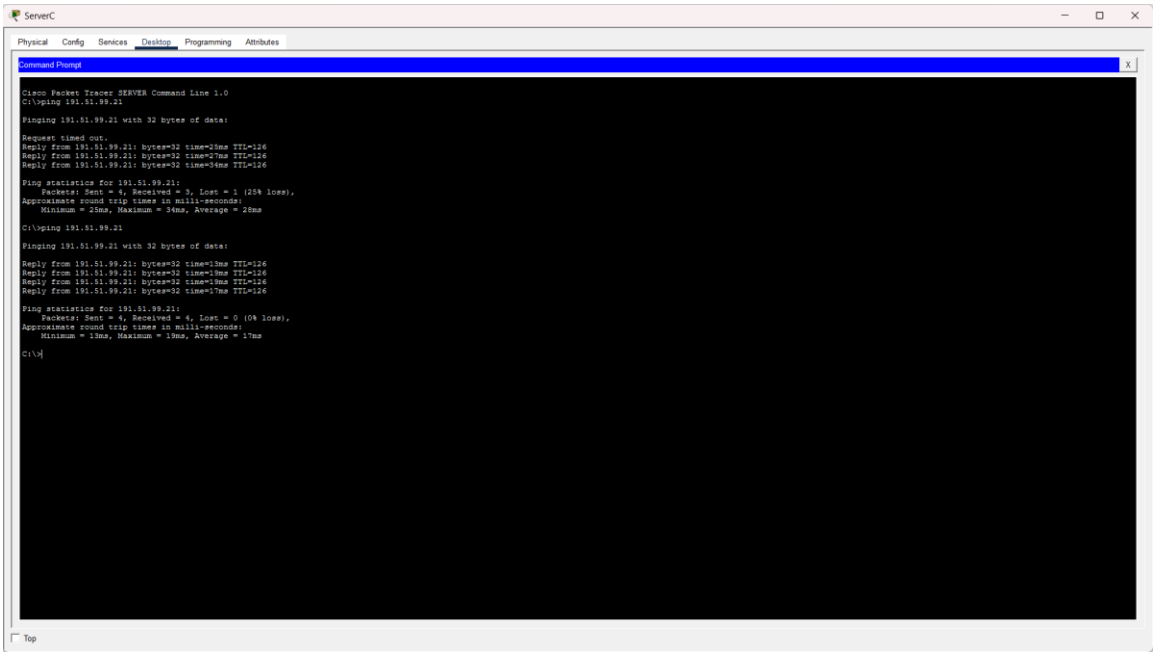
ServerC
Physical
Config
Services
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 191.51.99.21
Pinging 191.51.99.21 with 32 bytes of data:
Request timed out.
Reply from 191.51.99.21: bytes=32 time=25ms TTL=126
Reply from 191.51.99.21: bytes=32 time=27ms TTL=126
Reply from 191.51.99.21: bytes=32 time=34ms TTL=126
Ping statistics for 191.51.99.21:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 25ms, Maximum = 34ms, Average = 28ms
C:\>ping 191.51.99.21
Pinging 191.51.99.21 with 32 bytes of data:
Reply from 191.51.99.21: bytes=32 time=13ms TTL=126
Reply from 191.51.99.21: bytes=32 time=19ms TTL=126
Reply from 191.51.99.21: bytes=32 time=19ms TTL=126
Reply from 191.51.99.21: bytes=32 time=17ms TTL=126
Ping statistics for 191.51.99.21:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 13ms, Maximum = 19ms, Average = 17ms
C:\>
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